

VINUK RANAWEERA

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SUMMARY

Data Analyst skilled in Python, SQL, R, Tableau, and machine learning for data-driven decision-making and predictive modeling. Analyzed 5M+ records to identify customer behavior trends and developed classification models achieving 87% recall for loan default risk. Experienced across the complete data analytics process, from defining business questions and preparing data to advanced analysis, visual storytelling, and sharing actionable insights.

SKILLS

Programming: Python (pandas, NumPy, scikit-learn, Matplotlib), R (ggplot2, dplyr), SQL

Analytics: Data Visualization, EDA, Statistical Analysis, Predictive Modeling, Machine Learning, Feature Engineering

Visualization/Tools: Tableau, Excel, Google Sheets, BigQuery, MySQL, Jupyter, RStudio, Microsoft Word/PowerPoint

EXPERIENCE

Full-Stack Web Developer | Self-Employed/Freelance

Jan 2016 - Present

- Built and maintained content updates for the American Statistical Association NJ Chapter website.
- Designed and queried relational databases to manage application data, perform data validation, and support dynamic web functionality.

Research Intern | BCC Geospatial Center of the CUNY CREST Institute

Jul 2021 - Aug 2022

- Processed large-scale geospatial imagery datasets using Google Earth Engine APIs, generating GeoTIFF outputs for vegetation monitoring and spatial data visualization.
- Developed Python and Google Earth Engine tutorials for researchers to efficiently analyze satellite imagery.

Teacher Assistant (Junior Video Game Design)

Jan 2017 - Apr 2017

The Department for Lifelong Learning, Wagner College

- Mentored 25 middle school students in creating 2D video games using Scratch & Unity.

EDUCATION & CERTIFICATIONS

New York University (NYU) Tandon School of Engineering

Master of Science in Computer Engineering

Expected May 2027

MIT Professional Education

Applied Data Science Program: Leveraging AI for Effective Decision-Making

Aug 2024

- Applied machine learning to optimize e-commerce product recommendations and analyze marketplace behavior.
- Ranked among the top 5 highest accuracy for building a classification model in the final hackathon challenge.

Grove School of Engineering, The City College of New York (CUNY)

Bachelor of Science in Computer Science

Dec 2023

- Honors:* Cum Laude, Dean's List
- Relevant Coursework:* Data Structures, Algorithms, Software Design, Assembly Programming, Discrete Math, Probability Theory, Numerical Analysis in Programming, Modeling Complex Systems, Machine Learning

Coursera

IBM iOS and Android Mobile App Developer Certificate

Aug 2025

Google Data Analytics Professional Certificate

Jan 2024

- Developed skills in data analysis and visualization using spreadsheets, SQL, R, and Tableau.

Java Programming: Solving Problems with Software by Duke University Certificate

Jul 2019

PROJECTS

Loan Default Predictions | Python, Machine Learning, Predictive Modeling | [Link to Project](#)

Jul 2024 - Aug 2024

- Developed and evaluated classification models to predict loan default risk, achieving 87% recall on test data.
- Improved performance through feature engineering and hyperparameter tuning to reduce false negatives.

Potential Customer Predictions | Python, EDA, Machine Learning | [Link to Project](#)

Jun 2024 - Jul 2024

- Cleaned and analyzed lead generation datasets to identify key factors associated with customer conversion.
- Built and optimized Decision Tree and Random Forest models to identify high-value prospects.

Cyclistic Bike Share Case Study | R, Data Analysis, Data Visualization | [Link to Project](#)

Jan 2024 - May 2024

- Analyzed over 5 million rider records to evaluate behavioral trends among casual riders and annual members.
- Delivered data-driven recommendations projected to increase membership subscriptions by 10–15%.